

20	B	The constant electric force on the electron is pointing vertically downwards, in the direction of increasing potential. $ E = \left -\frac{\Delta V}{\Delta x} \right = \frac{2V}{d} \Rightarrow \Delta V = E (\Delta x) = \frac{2V}{d}(\Delta x) \text{ where } \Delta x \text{ is the vertical distance}$ work done by external force, $W = q(\Delta V) = -e(V_f - V_i)$ From X to Y: This is along an equipotential line. Since there is no change in potential, the work done is zero. From Y to Z: decrease in potential energy $W_{YZ} = (-e)(V_Z - V_Y) = (-e)\left(\frac{2V}{d}(\Delta x)\right) = -\frac{2V}{d}er \cos 30^\circ = -\frac{2V}{d}er \sin 60^\circ$ From Z to X: increase in potential energy $W_{ZX} = (-e)(V_X - V_Z) = (-e)\left(-\frac{2V}{d}(\Delta x)\right) = \frac{2V}{d}er \cos 30^\circ = \frac{2V}{d}er \sin 60^\circ$
21	B	increase in $K.E.$ = decrease in $P.E.$